

TRANSMITTAL	TRANSMITTAL NO. KC-RES-05-T01	DATE October 2026	SENT VIA Link · PDF attachment	SHEETS 10 sheets + 8 drawings
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TO Hiring team · MEP / BIM group, Metro Vancouver	FROM Kenneth Chow · Burnaby, BC · +1 778-791-3768 · wmchow5@gmail.com
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KENNETH CHOW

Re: MEP technician / MEP modeler roles, and coordinator roles where they fit. Enclosed is my BCIT residential MEP model and the code review I ran on it, item by item.

ITEMS ENCLOSED

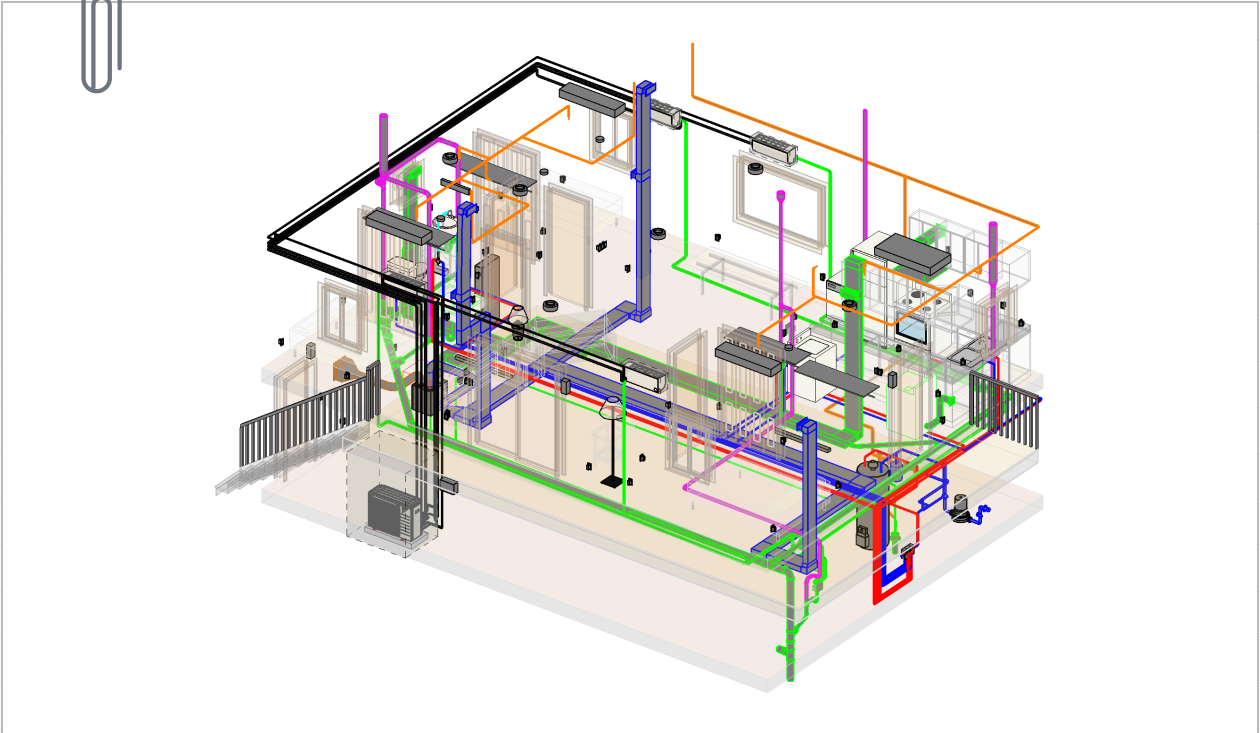
ENCL.	DESCRIPTION
01	3D MEP coordination model, KC-RES-05 (this page)
02	Applicant qualifications and model register
03	RFI log: 50 issues raised, 50 closed, 1 reopened on re-check
04	Six RFIs in full, with code basis and before / after
05	Eight drawings exported from the live model

ACTION REQUESTED

- ☒ Review the model and the decisions
- ☒ Invite to interview
- ☐ Revise and resubmit (tell me what to fix)

ENCL. 01 · 3D MEP coordination · view C-3D-OVERVIEW

Exported from Revit 2027, 2 Oct 2026



REMARKS

One-storey house on a basement, Burnaby BC. HVAC, plumbing, fire protection and electrical in one Revit model: 391 pipes (about 451 m), 26 circuits, 28 view templates, 8 sheets.

EMAIL wmchow5@gmail.com

PHONE +1 778-791-3768

APPLICANT QUALIFICATIONS		ENCL. 02	APPLICANT Kenneth Chow	BASED IN Burnaby, BC	
TRAINING Building Information Modeling (BIM) studies, BCIT, 2026. Bachelor of Engineering (Honours), Electronic and Communications, Hong Kong, 2015.	MODEL REGISTER · KC-RES-05				
	PIPES		391 · about 451 m		
	ELECTRICAL CIRCUITS		26 on panel P1 (200 A)		
EXPERIENCE Project Coordinator & Electronic Engineer (ELV systems), Shun Hing Engineering Contracting, Hong Kong, Nov 2020 - Sep 2023. Layout drawings, riser diagrams and shop drawings for access control, CCTV and intercom; RFIs, submittals and design markups; wall-cavity checks and power load calculations.	VIEW TEMPLATES / FILTERS		28 / 13		
	SHEETS		8		
	REVIT WARNINGS		2 (both from placeholder families)		
	ISSUES RAISED / CLOSED		50 / 50		
WHAT THAT MEANS HERE I have run the paperwork this page is built from. On KC-RES-05 I applied the same habit to my own model: raise the issue, cite the code, propose options, record the decision.	KEPT BY DESIGN (COURSEWORK)		3		
	REOPENED ON RE-CHECK		1 (SD1, see R-01)		
	BACKUPS BEFORE EACH CHANGE		one per item		
CODES WORKED TO ON THIS MODEL BC Building Code 2024 Part 9 · BC Plumbing Code · Canadian Electrical Code (as adopted in BC) · NFPA 13D · CSA F280, F326, B52 · CAN/ULC-S553 · Burnaby waterworks bylaw.	COURSEWORK NOTE The Daikin equipment, the dishwasher location and the sprinkler flows come from the BCIT brief. Where I changed coursework equipment (heat-pump heads, M-09) the log says so.				
SOFTWARE Autodesk Revit (MEP: piping, ducts, circuits, panel schedules, view templates, families) · AutoCAD · Revit API checks through pyRevit. Not yet used: Navisworks, ACC.					
HOW THE REVIEW WAS DONE The geometry and code checks were scripted through the Revit API with an AI assistant (Claude). I set the rules, chose every option and checked each fix in the model. I can walk through any item on this page.					

RFI LOG		ENCL. 03	PROJECT KC-RES-05	RAISED BY K. Chow (self-review)	PERIOD 28 Sep - 2 Oct 2026
51 rows					
RFI	DISC.	ISSUE	CODE BASIS	RESOLUTION	STATUS
C-01	HVAC	No heating: the only outdoor unit was cooling-only	BCBC 9.33 (22 °C)	Heat-pump outdoor unit with FTX heads; toe-kick heater in the bath	CLOSED · PLACEHOLDER FAMILY
C-02	FP	Sprinkler SP-07 hidden above the new hall ceiling	NFPA 13D coverage	Sidewall head lowered to 115 mm below the ceiling	CLOSED
C-03	FP	Fire main tees straight off domestic water with no components	NFPA 13D · CSA B64.10	Control valve, backflow, gauge, flow switch, drain & test	CLOSED · PLACEHOLDER FAMILY
C-04	ARCH	Deck guards 914 mm and open over a 2.1 m drop	BCBC 9.8.8	1070 mm guard, balusters 92 mm clear	CLOSED
C-05	ARCH	Exterior stair 610 mm wide, 203 mm run, open risers	BCBC 9.8	12 × 178 / 11 × 255, 934 mm clear, closed risers	CLOSED
C-06	ELEC	Bathroom switches inside the shower footprint	CEC · 500 mm from shower	Light switches on the hall side; fan timer 0.91 m from the shower	CLOSED
M-01 <u>FULL RFI</u>	PL	About 15 drains at 1 % and 4 at 0 %	BCPC 2.4.8.1	All drains ≤ 3" at 2 %; condensate also at 2 %	CLOSED
M-02	PL	Shower had no supply; a tempering valve fed nothing	CSA B125.3 · 49 °C	Pressure-balancing shower valve at 1.15 m	CLOSED · PLACEHOLDER FAMILY
M-03	HVAC	No range hood; HRV exhaust pickup over the range	BCBC 9.32 (~50 L/s)	150 cfm ducted hood; pickup moved 1.62 m away	CLOSED · PLACEHOLDER FAMILY
M-04	HVAC	No dryer exhaust	BCBC 9.32	100 mm rigid duct, connected to the dryer, to a wall hood	CLOSED
M-05	HVAC	Bathroom exhaust only 40 cfm	BCBC 9.32 (25 L/s)	55 cfm through the HRV boost timer	CLOSED
M-06	FA	No smoke alarm outside Bedroom 3	BCBC 9.10.19	SD6 on the vaulted ceiling, 0.82 m from the peak	CLOSED
M-07	FA	SD1 0.78 m from the AC-01 discharge	CAN/ULC-S553	Moved to 1.70 m from AC-01	CLOSED
M-08	FA	Smoke alarms reported as not circuited	CEC Section 32	Already on lighting circuits, none on GFCI	CLOSED
M-09 <u>FULL RFI</u>	HVAC	No F280 calculation; living-room head undersized	CSA F280	FTX15 × 3 + FTX35 heat-pump heads (coursework deviation)	CLOSED · PLACEHOLDER FAMILY
M-10 <u>FULL RFI</u>	HVAC	R32 charge against the smallest room	CSA B52 / IEC 60335-2-40	2.5 kg charge ≤ 2.80 kg allowed	CLOSED
M-11	PL	Water heater missing T&P valve, expansion tank, straps	BCPC	T&P valve, ¾" copper discharge, 2 gal tank; straps noted	CLOSED
M-12 <u>FULL RFI</u>	PL	PVC pipe running through the water heater tank; no floor drain	BCPC traps	Floor drain with P-trap (95 mm seal); stack extended	CLOSED
M-13	ARCH	East deck stair missing	BCBC 9.8	5 × 173 / 4 × 280, 1.30 m wide, guards both sides	CLOSED
V-01	ARCH	Basement door 1930 mm high	BCBC Table 9.5.5.1	813 × 1981 door, 187 mm left for the header	CLOSED
V-02	PL	Water service only 0.26 m below grade	Burnaby bylaw 762 mm	Entry lowered to -1.50 m, about 1.0 m cover	CLOSED
V-03	PL	No cleanout and no sewer connection	BCPC	Cleanout inside on the stack; 4" sewer at 1 %	CLOSED
V-04	ARCH	Bedroom egress relies on the sprinkler waiver	BCBC 9.9.10.1	Dependency documented on the windows	DOCUMENTED
V-05	PL	Condensate lines 16 mm	Daikin (VP20 / ¾")	All four lines upsized to ¾"	CLOSED
P-01	HVAC	AC-01 cuts into the bedroom ceiling	Daikin ≥ 50 mm top	Lowered 130 mm; line sets kept clear of a vent	CLOSED
P-02	HVAC	AC-02 cuts into the ceiling	Daikin ≥ 50 mm top	Lowered 130 mm	CLOSED
P-03	HVAC	AC-03 cuts into the ceiling	Daikin ≥ 50 mm top	Lowered 130 mm	CLOSED
P-04	HVAC	Outdoor unit embedded in the wall, under the deck	Daikin clearances	Pulled out 100 mm onto a 100 mm pad	CLOSED

RFI	DISC.	ISSUE	CODE BASIS	RESOLUTION	STATUS
P-05	ELEC	Panel P1 runs into the floor joists	CEC 2-308 working space	Moved to the west basement wall, 1 m clear	CLOSED
P-06	HVAC	HRV intake labelled return air, exhaust as supply	System classification	New Outdoor Air and Exhaust Discharge system types	CLOSED
P-07	HVAC	Six 5 cfm token exhaust grilles	CSA F326 room rates	Removed; exhaust moved to bath, kitchen, laundry	CLOSED
P-08	HVAC	Supply air into the laundry	CSA F326	Removed; master bedroom raised to 22 cfm	CLOSED
P-09 FULL RFI	HVAC	HRV exhaust under the deck, near the outdoor unit	HRV manual	HRV rotated and moved; grilles on the west wall 2.47 m apart	CLOSED
P-10	HVAC	No HRV condensate drain	BCPC air gap ≥ 25 mm	Condensate pump to the receptor, 26 mm air gap	CLOSED
P-11	PL	Washer piped rigidly from its base	BCPC traps and air break	Washer box, 2" standpipe and trap	CLOSED
P-12	PL	No main shut-off, check or isolation valves	BCPC (PRV > 550 kPa)	Shut-off, check and isolation valves added	CLOSED · PLACEHOLDER FAMILY
P-13	PL	Dishwasher 3.4 m from the sink	BCPC trap and vent	Kept: coursework location; drain checked compliant	KEPT BY DESIGN
P-14	FP	Sprinkler flows 32-43 gpm per head	NFPA 13D minimums	Kept: coursework values, conservative	KEPT BY DESIGN
P-15	FP	Fire pipe type was the generic "Standard"	NFPA 13D (copper K)	Own Copper Type K fire-protection type	CLOSED
P-16	FP	Fire pipes above ceilings may freeze	NFPA 13D	Warm-side installation notes	DOCUMENTED
P-17	FP	Light L1 50 mm from sprinkler SP-01	13D obstruction (24")	Light moved 0.6 m; 0.65 m clear	CLOSED
P-18	ELEC	Lights floating in vaulted rooms	Model accuracy	Pendant types with stem lengths	CLOSED
P-19	ELEC	Receptacle R25 60 mm from the wall end	100 mm from jamb or end	Moved with switch S9; 115 mm	CLOSED
P-20	ELEC	Switches hidden in the laundry alcove	Latch-side placement	Moved to the alcove wall; hinge side flagged	CLOSED
P-21 FULL RFI	ELEC	100 A service with little margin	CEC 8-200	200 A service; 62 % with a future EV	CLOSED
P-22	ELEC	No AFCI, weather-resistant or battery-backup specs	CEC 26-658 · BCBC 9.10.19	Specs on every circuit and device type	CLOSED
P-23	BIM	Confusing level names; elements without a level	BIM standard	Levels renamed; 117 schedule levels set	CLOSED
P-24	BIM	Metric family names in an imperial project	BIM standard	Renamed to imperial	CLOSED
P-25	BIM	Window type names didn't match sizes	BIM standard	Renamed to real sizes	CLOSED
P-26	BIM	Architecture and MEP in one file, no links	BIM practice	Kept: coursework single file; split plan written	KEPT BY DESIGN
R-01	FA	SD1 about 0.64 m from supply diffuser AT-06	CAN/ULC-S553 (~0.9 m)	Found on re-check; move to about (-3.3, 3.4)	OPEN

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REQUEST FOR INFORMATION

RFI NO.
C-03

ENCL.
04-1

DISCIPLINE
Fire protection

STATUS
Closed

SUBJECT **Fire sprinkler main fed straight off domestic water**

QUESTION

The 1¼" sprinkler main teed off the domestic cold water through a transition fitting. No backflow preventer, control valve, gauge or drain & test.

OPTIONS CONSIDERED

1. Add the standard 13D riser components. 2. Treat it as a multipurpose system. Rejected: the sprinkler piping dead-ends, so water never circulates.

RESPONSE

Five in-line placeholders on the fire main in the basement, with the drain & test discharging over the floor drain. Then I swapped two of them (see after).

CODE BASIS

NFPA 13D riser components and waterflow alarm · BC Plumbing Code backflow, CSA B64.10 (type set by the City).

MY CALL

Asking what the drain pipe was for exposed the real error: test water left at the drain & test before reaching the flow switch.

REFERENCE

FP-CV1 1774562 · FP-BFP1 1774565 · FP-PG1 1774568 · FP-FS1 1774574 · FP-DT1 1774571

BEFORE

DCW tee → fire main → sprinklers

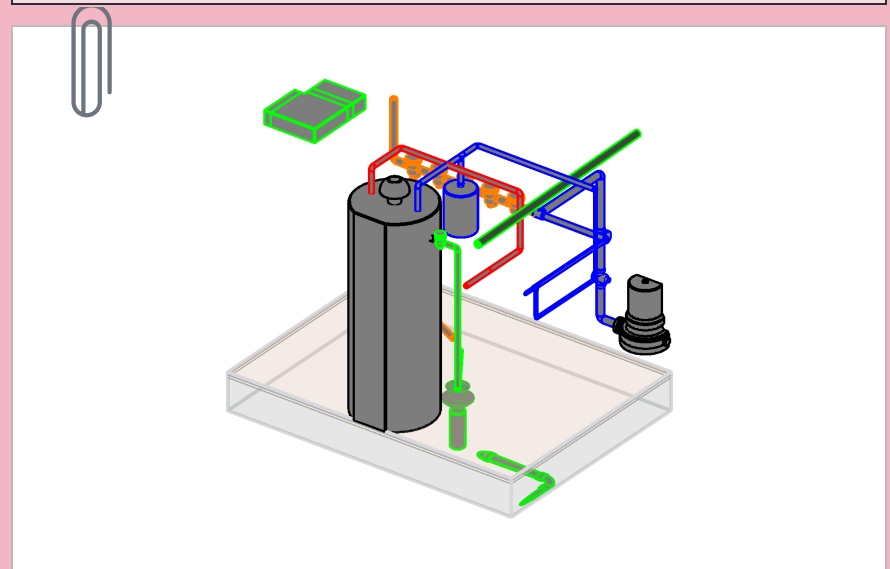
No way to isolate, test, or stop stagnant sprinkler water flowing back into drinking water.

First build order: CV → BFP → PG → D&T → FS. A flow test would never pass the switch.

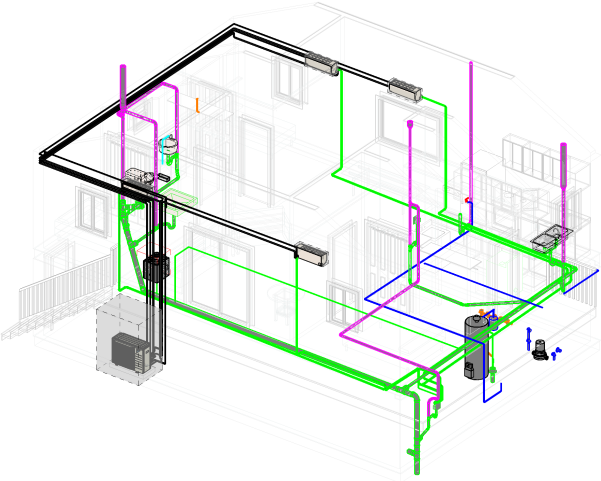
AFTER

DCW tee → CV → BFP → PG → FS → D&T → up

Opening the drain & test now sends water through the flow switch first, so the test proves the alarm.



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REQUEST FOR INFORMATION	RFI NO. M-01	ENCL. 04-2	DISCIPLINE Plumbing	STATUS Closed
SUBJECT Drainage at 1 % and 0 % slope				
QUESTION About 15 horizontal 1½" and 2" drains were at 1 %, four at 0 %, and the four condensate lines at 1 %.		BEFORE 2" branch Condensate: 16 mm lines at 1 %. 1 % · 4 at 0 % AFTER 2" branch Exit invert unchanged at the time; 4" building drain stays at 1 %. Clash check clean. 2 % (¼" per ft) 		
OPTIONS CONSIDERED Re-slope sanitary only and leave condensate at 1 % (common HVAC practice), or follow the code for both.				
RESPONSE Every drain 3" and under now at 2 %, upstream ends raised so the 4" building drain and its exit stay put. Condensate at 2 % with five new corner drops to clear crossings.				
CODE BASIS BC Plumbing Code 2.4.8.1: drains ≤ 3" at least 1 in 50 (2 %); over 3" may be 1 in 100. Condensate to a receptor is indirect waste, which is drainage pipe.				
MY CALL <i>I asked: "under BC law, is 1 % fine? If not, follow the law."</i>				
REFERENCE Kitchen and DW main · KS-01 arm · washer branch · shower · condensate IW1-IW4				

REQUEST FOR INFORMATION

RFI NO.
M-09 · M-10

ENCL.
04-3

DISCIPLINE
HVAC

STATUS
Closed

SUBJECT **Heat-pump sizing and the R32 charge limit**

QUESTION

No CSA F280 calculation. The living room's 2.0 kW head served a 35 m² room with a 5.9 m vault. The heads were cooling-only, so the house had no real heat.

OPTIONS CONSIDERED

A. Heat-pump FTX heads on a heat-pump outdoor unit. B. Keep the coursework cooling-only heads and heat with about 2.3 kW of extra baseboard, which pushes the panel past 100 A.

RESPONSE

Option A, recorded as a deviation from the BCIT equipment spec (FTC15NV14 × 3 + FTC20NV14). Then checked the R32 charge against the smallest room.

CODE BASIS

BCBC 9.33 heating to CSA F280 · CSA B52 / IEC 60335-2-40 charge limit for A2L refrigerant.

MY CALL

"The AC units are a coursework requirement... but I know option A is more efficient. Go ahead anyway."

REFERENCE

AC-01 1524859 · AC-02 1525077 · AC-03 1525454 · AC-04 1526631 · CU-01 1527078

BEFORE

AC-04 FTC20: **2.0 kW** against an open-zone cooling load of **3.3 kW**.

All four heads FTC: cooling only. Heating came from 500 W baseboards.

AFTER

F280-STYLE ESTIMATE house 3.6 kW heat / 4.6 kW cool

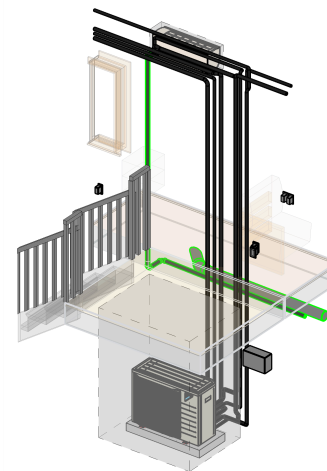
HEADS FTX15 × 3 + FTX35

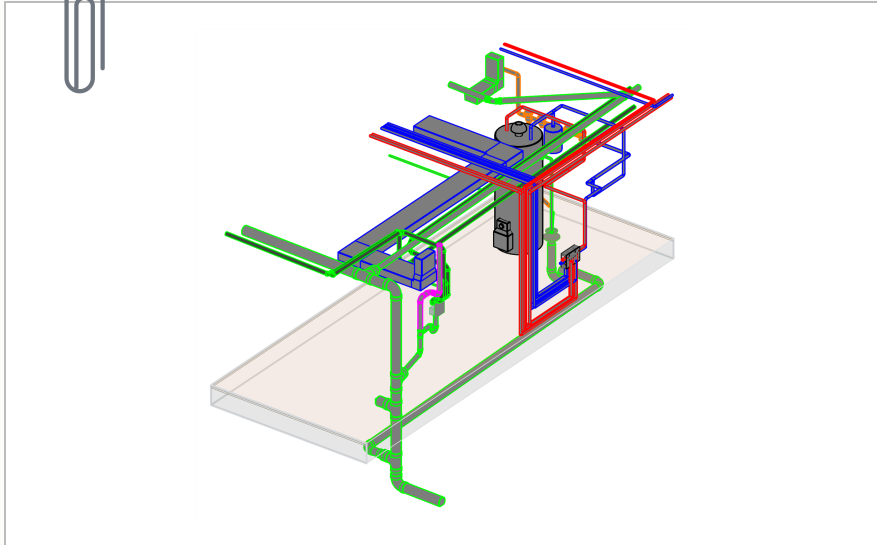
LIQUID LINES 18.9 + 5.8 + 8.9 + 22.3 = 55.9 m

CHARGE 2.0 + (55.9 - 30) × 0.02 = **2.5 kg**

m_{max} (Bedroom 1) 2.5 × 0.307^{1.25} × 1.8 × √7.44 = **2.80 kg**

RESULT 2.5 ≤ 2.80, passes with no mitigation



REQUEST FOR INFORMATION		RFI NO. M-12	ENCL. 04-4	DISCIPLINE Plumbing	STATUS Closed
SUBJECT A pipe through the water heater, and no floor drain					
QUESTION A 1" PVC pipe hung off the water heater's drain-valve connector with its centreline inside the tank wall. Its comment said it was the T&P discharge. The basement had no floor drain for it.			BEFORE 1" PVC on the tank's drain valve, clipping the tank. PVC is not rated for 99 °C relief water. No floor drain; building drain exit only 0.14 m below the floor.		
OPTIONS CONSIDERED (a) Extend the building drain stack down and drain a floor drain by gravity. (b) A sump and pump.			AFTER WH-01 → TPV-01 → ¾" Cu → air gap → FD-01 → P-trap → stack Trap seal about 95 mm; trap seal primer noted on FD-01.		
RESPONSE Deleted the pipe; added a real T&P valve with a ¾" copper discharge; floor drain FD-01 with a P-trap, 3" drain at 2 % to a new sanitary tee; stack extended from -2.19 to -2.95 m.					
CODE BASIS BC Plumbing Code: T&P discharge full size and temperature-rated, no valves; every fixture trapped, seal ≥ 38 mm; floor drains need trap seal protection.					
MY CALL <i>I rejected two trap placements, then placed the trap myself the right way up and had the method written into a reusable skill.</i>					
REFERENCE FD-01 1782449 · trap 1783819 · tee 1782637 · TPV-01 1782320 · WH-01 1602886					

REQUEST FOR INFORMATION

RFI NO.
P-09

ENCL.
04-5

DISCIPLINE
HVAC

STATUS
Closed

SUBJECT **HRV exhaust discharging under the deck**

QUESTION

The HRV exhaust grille sat under the deck, 0.53 m from the outdoor heat pump's clearance zone.

OPTIONS CONSIDERED

Keep it · slide it east on the south wall · run a long duct to the west wall · rotate the HRV · rotate and move the HRV. Compared in one table.

RESPONSE

Rotated HRV-01 90° and moved it beside the west wall: both grilles on the west wall, 2.47 m apart, out from under the deck. Then added the receptacle and condensate pump it was missing.

CODE BASIS

HRV manufacturer separation (≥ 1.8 m) and height above grade (≥ 450 mm); keep moist exhaust off deck framing and the heat-pump coil.

MY CALL

The discharge duct looked too long, so I suggested rotating the HRV, then asked about its power socket.

REFERENCE

HRV-01 1547124 · AT-10 1567942 · AT-17 1785468 · REC-HRV 1785491 · CP-01 1788255

BEFORE

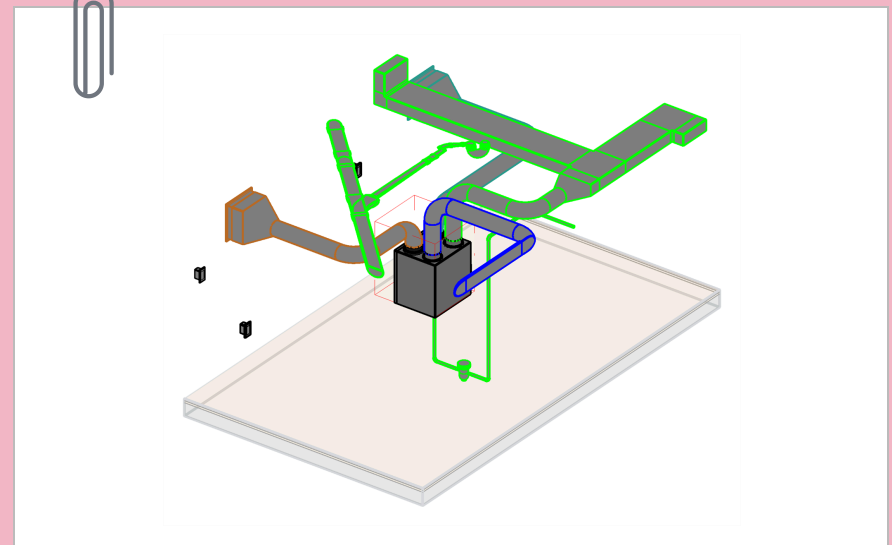
Exhaust under the deck, **0.53 m** from the heat-pump clearance zone. Moist air on deck joists, frost on the coil.

Outdoor ducting about 4.5 m. No receptacle and no condensate drain at the HRV.

AFTER

Grilles on the west wall, **2.47 m** apart; discharge about 3.5 m from the heat pump.

Outdoor ducting about 3.0 m straight. Dedicated receptacle REC-HRV; condensate pump CP-01 to the receptor with a 26 mm air gap.



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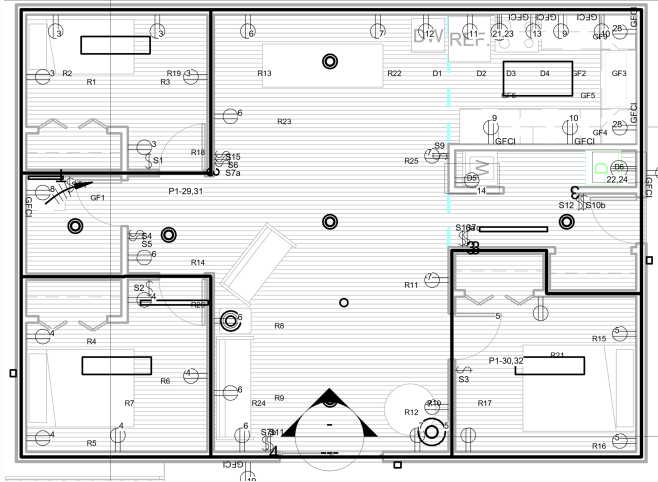
DRAWING ENCLOSURES

ENCL.
05 A-H

SOURCE
Revit views, exported 2
Oct 2026

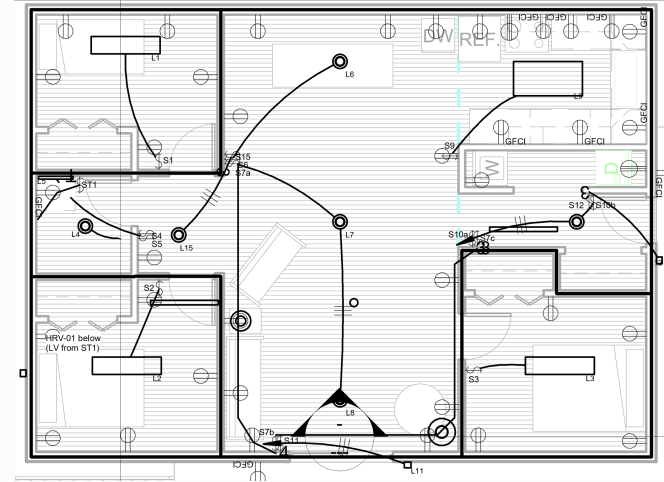
NOTE
Title blocks cropped

ENCL. 05-A · Power plan, Level 1

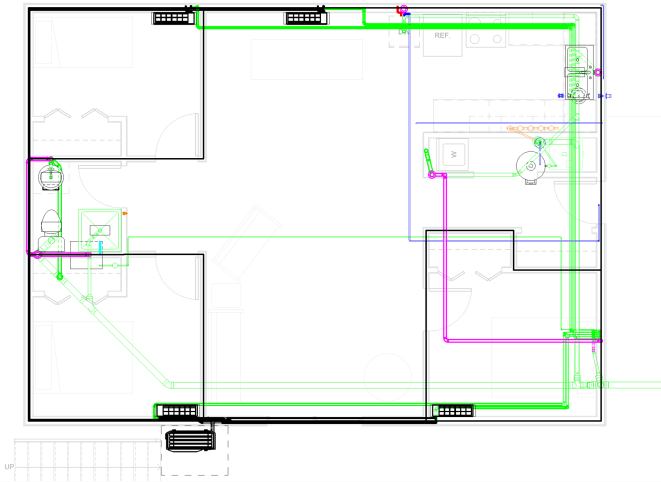


ENCL. 05-A · POWER PLAN, LEVEL 1

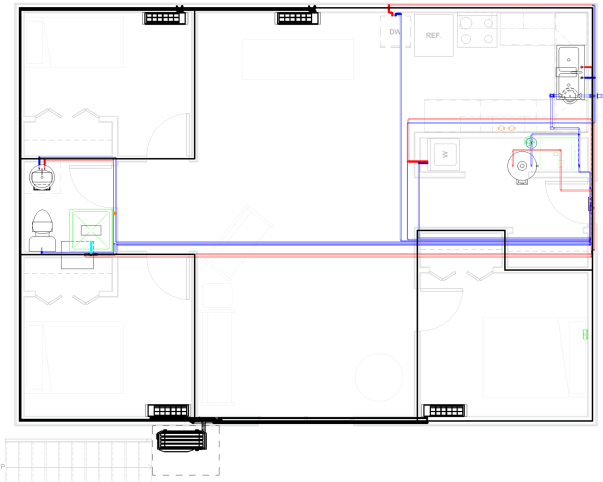
[Click the drawing to enlarge](#)



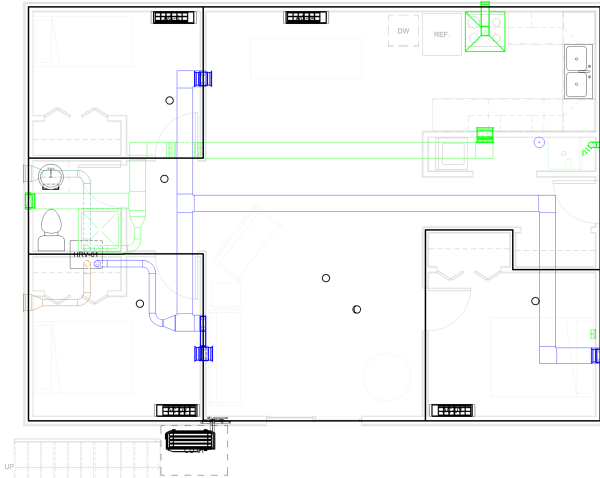
ENCL. 05-B · LIGHTING PLAN, LEVEL 1



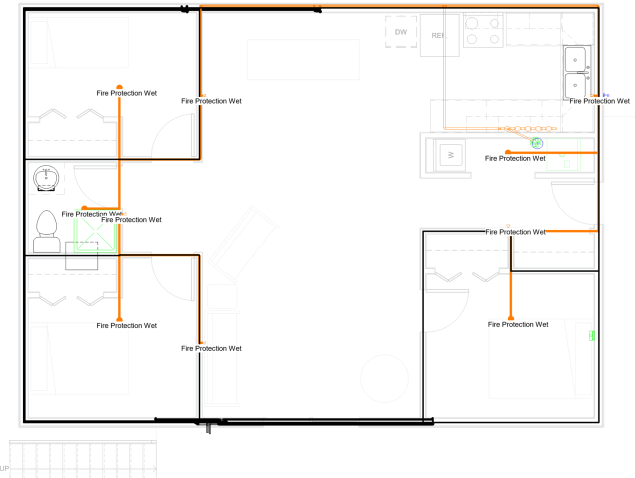
ENCL. 05-C · SANITARY DRAINAGE AND VENT



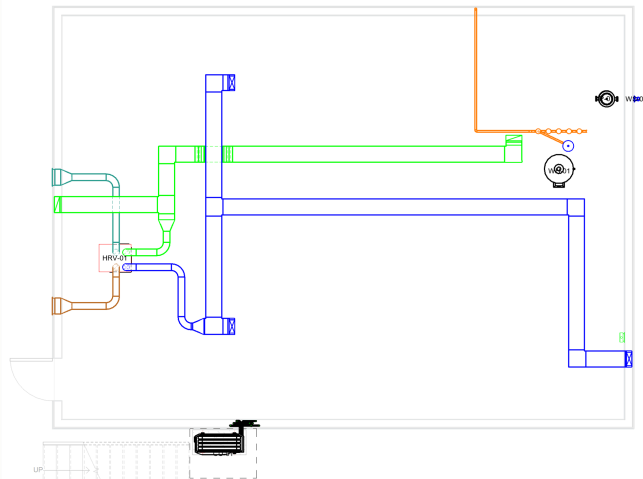
ENCL. 05-D · DOMESTIC WATER



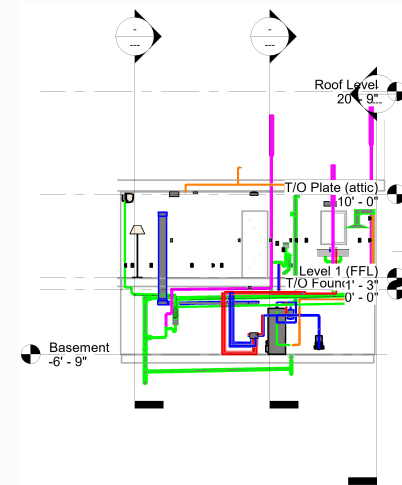
ENCL. 05-E · HVAC, LEVEL 1



ENCL. 05-F · FIRE SPRINKLERS, LEVEL 1



ENCL. 05-G · BASEMENT PLANT



ENCL. 05-H · COORDINATION SECTION 01

REVIEWER ACTION

TRANSMITTAL NO.
KC-RES-05-T01

RETURN TO
Kenneth Chow

EMAIL **wmchow5@gmail.com**

PHONE **+1 778-791-3768**

BASED IN **Burnaby, BC · available for Metro Vancouver**

OPEN ITEMS CARRIED FORWARD

R-01: smoke alarm SD1 sits about 0.64 m from supply diffuser AT-06; it needs about 0.9 m. Found on my own re-check after the log closed. Fix: move SD1 to about (-3.3, 3.4).

Placeholder families (heat pump, valves, shower valve, range hood) stand in for manufacturer families. They cause the 2 remaining Revit warnings.

PREPARED BY

Kenneth Chow · BIM & design coordination · BCIT BIM studies 2026.
Model review assisted by Claude through the Revit API; every decision is mine and I can explain it.

~~CANARY - RETURN COPY~~